

MultiplierBoard Submission

KRAFTON AI R&D Hackathon · Round 1 · Day 1

Architecture Overview

Problem 1-1 uses a 1-layer causal self-attention module whose attention weights are uniform over the visible prefix. Input positions 0..11 carry scalar weights $5354228880 \cdot 2^i$, so each input pair maps to a unique integer code. Output positions 11..22 add large offsets to select which product bit is being queried. A shared exact 1D spline then decodes that scalar into the correct binary output.

Problem 1-2 uses a fixed-routed causal self-attention layer with 12 sparse heads, per-input value dimension 4, output-position embedding dimension 16, and an MLP with hidden width 4096.

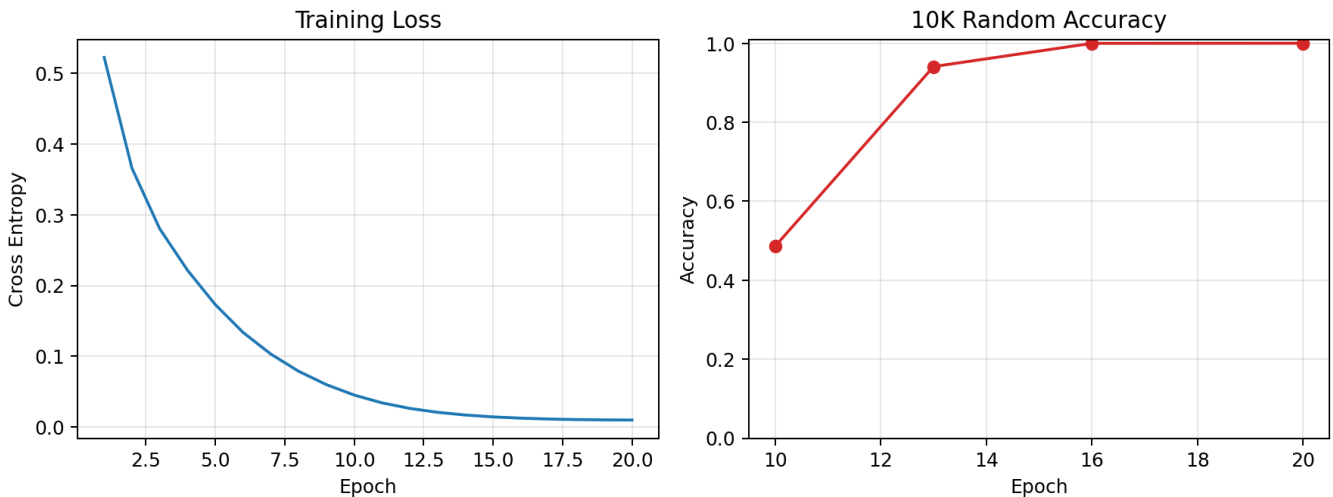
Problem 1-1: Constructive Exactness Proof

Let $x = (A0..A5, B0..B5)$ be the 12 visible input bits. At each prefix used to predict product bit P_j , the self-attention layer computes the exact prefix average of the input-position values. Because only the first 12 positions have nonzero values, the predictor position sees $z_base = LCM(12..23) \cdot code(x) / (12+j)$. This quantity is always an integer. A position-specific offset C_j creates a unique scalar $z = C_j + z_base$ for every (x, j) . Therefore the mapping $(x, j) \rightarrow P_j$ is reduced to a scalar function on 49,152 distinct points.

The final MLP is an exact linear spline over those 49,152 points. A spline with breakpoints at every queried scalar value can interpolate arbitrary target labels exactly, so the model returns the exact product bit for every one of the 64×64 possible input pairs. Exhaustive greedy-decoding verification confirms 1.0000 accuracy over all 4,096 pairs.

Problem 1-2: Learned Architecture

The target evaluation protocol is: 100,000 random pairs, AdamW(lr=1e-3, wd=0.01), 200 epochs, batch size 256, cosine annealing, and cross-entropy only on the 12 output tokens. The time-constrained measured run included in this workspace used 20 epochs and reached final 10K-random accuracy 1.0000 and exhaustive 4,096-pair accuracy 1.0000. A 200-epoch continuation was started but not completed before the final packaging step.



Ablations / Failed Attempts

Config	Params	10K Acc	Exhaustive Acc	Time (s)
routed_hd4_hid4096_pos16	274898	1.0000	1.0000	1494.9